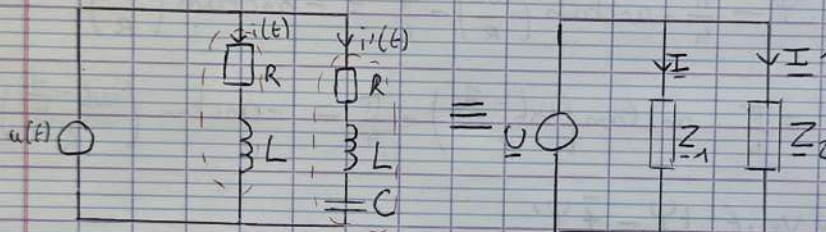


Exercice 6.

$$u(t) = U_M \cos(\omega t + \frac{\pi}{4})$$

Grandeurs temporelles	mb complexe	Ampl. complexe
$u(t) = U_M \cos(\omega t + \frac{\pi}{4})$	$\underline{u}(t) = \underline{U} e^{j\omega t}$	$\underline{U} = U_M e^{j\frac{\pi}{4}}$
$i(t) = I_M \cos(\omega t + \varphi)$	$\underline{i}(t) = \underline{I} e^{j\omega t}$	$\underline{I} = I_M e^{j\varphi}$
$i'(t) = I'_M \cos(\omega t + \varphi')$	$\underline{i}'(t) = \underline{I}' e^{j\omega t}$	$\underline{I}' = I'_M e^{j\varphi'}$



Association en série: $\underline{Z}_1 = \underline{Z}_L + R = j\omega L + R$ $\underline{Z}_2 = \underline{Z}_L + R + \underline{Z}_C = j\omega L + R + \frac{1}{j\omega C}$

Loi d'ohm complexe: $\underline{U} = \underline{Z}_1 \underline{I} \Leftrightarrow \underline{I} = \frac{\underline{U}}{\underline{Z}_1}$

$$|\underline{I}| = \frac{|\underline{U}|}{|\underline{Z}_1|} = \frac{U_M}{\sqrt{\omega^2 L^2 + R^2}} \quad \Leftrightarrow \quad \underline{I} = \frac{\underline{U}}{(j\omega L + R)}$$

$$\varphi = \arg(\underline{U}) - \arg(\underline{Z}_1) = \frac{\pi}{4} - \arctan\left(\frac{\omega L}{R}\right)$$

$$\underline{I} = \frac{U_M}{\sqrt{\omega^2 L^2 + R^2}} e^{j(\frac{\pi}{4} - \arctan(\frac{\omega L}{R}))}$$

$$i(t) = \frac{U_M}{\sqrt{\omega^2 L^2 + R^2}} \cos\left(\omega t + \frac{\pi}{4} - \arctan\left(\frac{\omega L}{R}\right)\right)$$

$\underline{U} = \underline{Z}_2 \underline{I}' \Leftrightarrow \underline{I}' = \frac{\underline{U}}{\underline{Z}_2}$ (loi d'ohm complexe)

$$\underline{I}' = \frac{\underline{U}}{j\omega L + R + \frac{1}{j\omega C}}$$

$$|I'| = \frac{|U|}{|Z_2|} = \frac{U_M}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$$

$$\begin{aligned} \arg(I') &= \arg(U) - \arg(Z_2) \\ &= \frac{\pi}{4} - \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) \end{aligned}$$

$$\text{donc } i'(t) = \frac{U_M}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \cos\left(\omega t + \frac{\pi}{4} - \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)\right)$$

$$2. \quad \Psi = \frac{\pi}{4} - \arctan\left(\frac{\omega L}{R}\right) - \frac{\pi}{4} = -\arctan\left(\frac{\omega L}{R}\right)$$

$$\Psi' = \frac{\pi}{4} - \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) - \frac{\pi}{4} = -\arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

$$\text{on veut } +\Psi = -\Psi'$$

$$\Rightarrow -\arctan\left(\frac{\omega L}{R}\right) = +\arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right) \quad \downarrow \text{ arctan impaire}$$

$$\Rightarrow \arctan\left(-\frac{\omega L}{R}\right) = \arctan\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$

$$\Rightarrow -\frac{\omega L}{R} = \frac{\omega L - \frac{1}{\omega C}}{R} \quad \checkmark \arctan(\arctan(x)) = x$$

$$\Rightarrow \frac{2\omega L - \frac{1}{\omega C}}{R} = 0$$

$$\Rightarrow 2\omega L - \frac{1}{\omega C} = 0$$

$$\Rightarrow 2\omega L = \frac{1}{\omega C}$$